Extensive-form games

Pure strategies

- Let $\Gamma = \langle N, V, E, x_0, (V_i)_{i \in N \cup \{0\}}, \mathcal{H} = (H_i)_{i \in N}, \mathcal{A} = (A(h))_{h \in \cup_{i \in N} H_i}, (p_x)_{x \in V_0}, (u_i)_{i \in N} \rangle$ be an extensive form game.

Definition

A pure strategy of s_i of Player i is a mapping assigning an action $a \in A(h)$ to each information set $h \in H_i$, i.e.,

$$s_i : H_i \to \cup_{h \in H_i} A(h) \text{ such that } s_i(h) \in A(h) \text{ for all } h \in H_i.$$

Pure strategies

- By S_i we denote the set of all pure strategies of Player i.
- A strategy profile $s = (s_1, \ldots, s_n)$ is a collection of strategies of all the players where n is the number of players.
- By $S = S_1 \times \cdots \times S_n$ we denote the set of all strategy profiles.

Example

- Consider the original battle of the sexes game

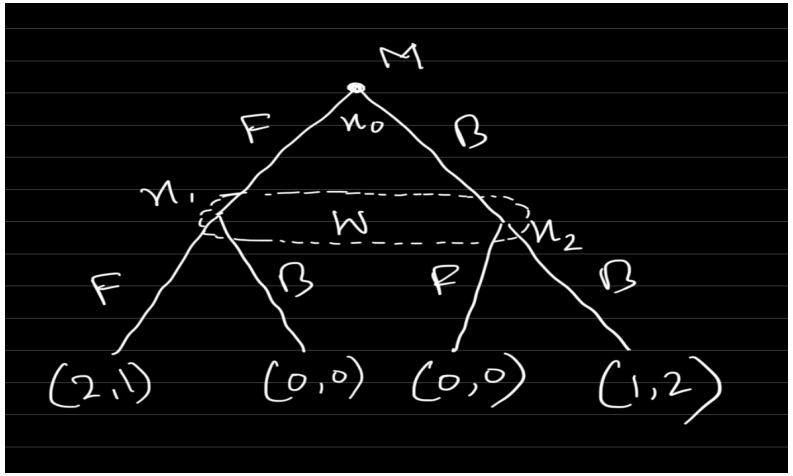

Figure: Figure 1

- Here Player M has one information set with one node and two strategies, i.e., $S_M = \{F, B\}$ where F represents the edge (x_0, x_1) and B represents the edge (x_0, x_2).

Example

- The Player W also has one information set but with two nodes and two strategies, $S_M = \{F, B\}$ where F represents the collection of edges $\{(x_1, x_3), (x_2, x_5)\}$ and B represents the collection of edges $\{(x_1, x_4), (x_2, x_6)\}$.

Mixed strategies

- As before, we define mixed strategies as probability distributions of pure strategies.
- Formally, a mixed strategy σ_i is a mapping from S_i to $[0, 1]$ such that $\sum_{s_i \in S_i} \sigma_i(s_i) = 1$.
- By Σ_i we denote the set of all pure strategies of Player i.
- A strategy profile $\sigma = (\sigma_1, \ldots, \sigma_n)$ is a collection of mixed strategies of all the players where n is the number of players.
- By $\Sigma = \Sigma_1 \times \cdots \times \Sigma_n$ we denote the set of all strategy profiles.

Example

- In the example of the battle of the sexes, both players have the same mixed strategy space Δ_2, where

$$\Delta_p = \{(x_1, \ldots, x_p) \mid x_i \geq 0 \text{ for all } i \text{ and } \sum_{i=1}^{p} x_i = 1\}.$$

Example

- Consider the following game

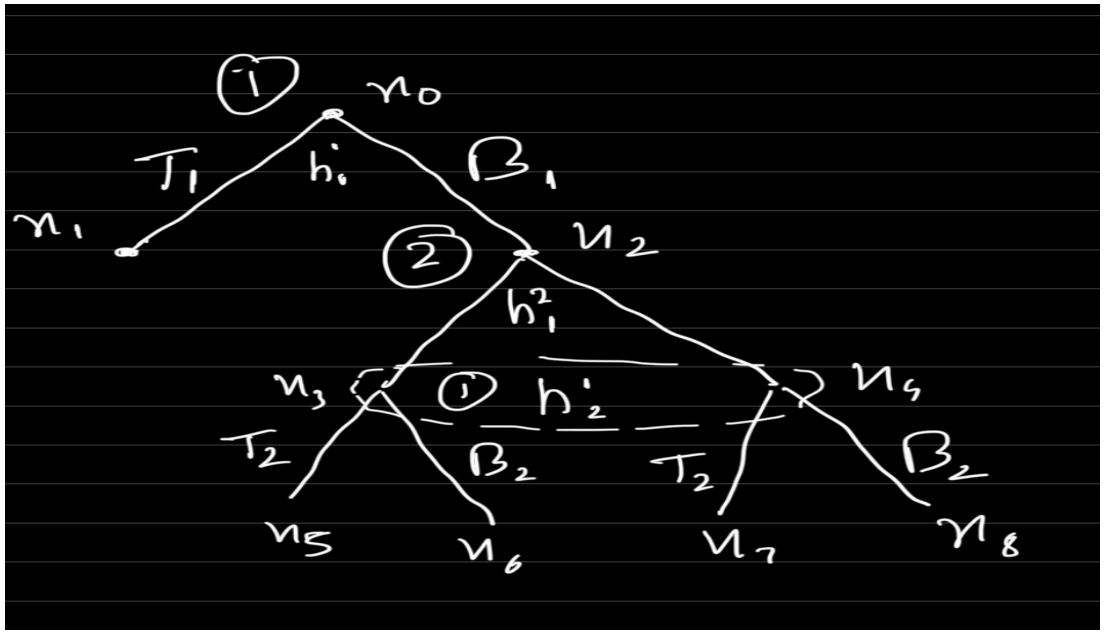

Figure: Figure 2

- Note that in this game, Player 1 has four pure strategies, $S_1 = \{T_1T_2, T_1B_2, B_1T_2, B_1B_2\}$.

Example

- Therefore, the mixed strategy space

$$\Sigma_1 = \Delta_4 = \{(x_1, \ldots, x_4) \mid x_i \geq 0 \text{ for all } i \text{ and } \sum_{i=1}^{4} x_i = 1\}.$$

- The mixed strategy space is equivalent to a subset of $\mathbb{R}^4$, which is three-dimensional.

Example

- However, Player 1 has another, equally natural, alternative way to attain randomness: he can choose randomly between T_1 and B_1 when the play of the game arrives at his information set h_1^1 , and then choose randomly between T_2 and B_2 when the play of the game arrives at his information set h_2^1.
- Such a strategy can be expressed in terms of two probability distributions $[\alpha(T_1), (1-\alpha)(B_1)]$ and $[\beta(T_2), (1-\beta)(B_2)]$

Example

- These strategies are called behavior strategies.
- Note that the set of all such strategies is equivalent to a subset of $\mathbb{R}^2$.

Behavior strategies

- A behavior strategy of a player in an extensive-form game is a function mapping each of his information sets to a probability distribution over the set of possible actions at that information set.
- Formally, for Player i, a behavior strategy b_i is a mapping from H_i to $\cup_{h \in H_i} \Delta A(h)$ such that $b_i(h) \in \Delta A(h)$ for all $h \in H_i$.
- A behavior strategy is a vector of probability distributions, one per information set.

Behavior strategies

- The probability that a behavior strategy b_i will choose an action $a \in A(h_i)$ at an information set h_i is denoted by $b_i(a; h_i)$.
- The set of all behavior strategies of Player i is denoted by $\mathcal{B}_i$.
- We write a behavior strategy profile as $b = (b_1, \ldots, b_n)$, and the set of all profiles by $\mathcal{B} = \times_{i \in N} \mathcal{B}_i$.
- Note that in the previous example, Σ_1 is three-dimensional whereas $\mathcal{B}_1$ is two-dimensional, which suggests that Σ_1 could be a larger set than $\mathcal{B}_1$.

Question!

- Note that both mixed strategies and behavior strategies capture the idea of randomly playing in a game.
- Therefore, it's a legitimate question to ask is there an essential difference between these two strategies? Can a player attain a higher utility by using a behavior strategy instead of a mixed strategy? Alternatively, can he attain a higher utility by using a mixed strategy instead of a behavior strategy?
- Also, what is the relationship between Σ_i and $\mathcal{B}_i$, in general?

Mixed/behavior strategy profile

- To define formally the equivalence between a mixed strategy and a behavior strategy, we consider strategy vectors that consist of both mixed strategies and behavior strategies.
- With a slight abuse of notations, we denote by $\sigma = (\sigma_i)_{i \in N}$, a strategy profile where σ_i can be either a mixed or a behavior strategy for Player i, for each i.
- For every mixed/behavior strategy vector $\sigma = (\sigma_i)_{i \in N}$, and every node x in the game tree, denote by $\rho(x; \sigma)$ the probability that node x will be visited during the course of the play of the game when the players implement strategies $\sigma = (\sigma_i)_{i \in N}$,.

Example

- Consider the following example of a game:

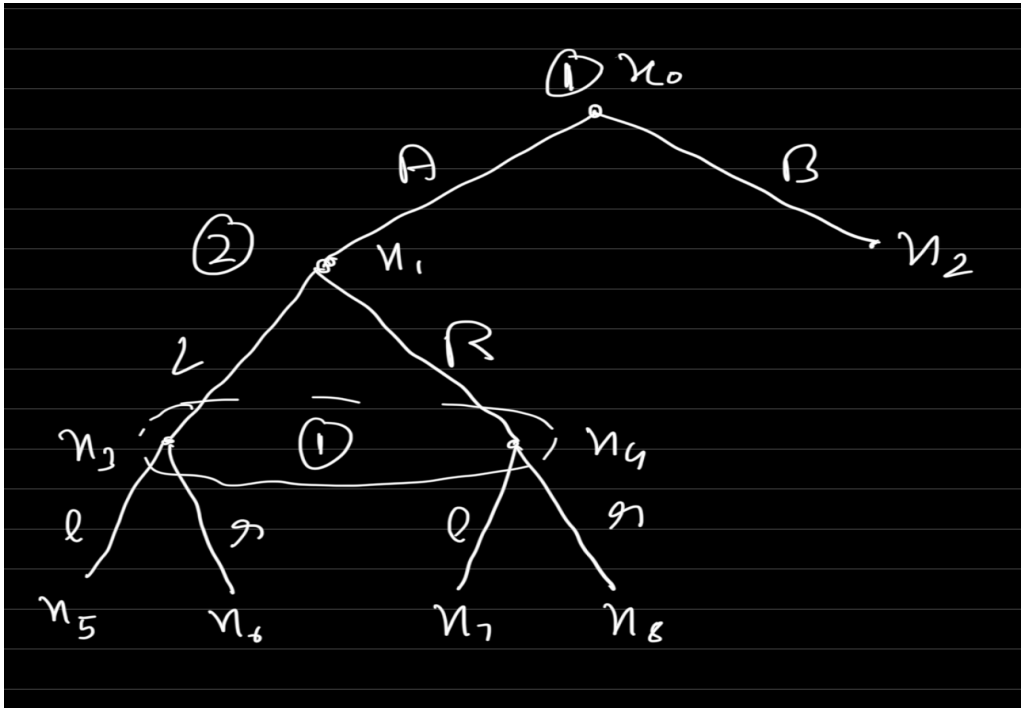

Figure: Figure 3

Example

- And consider the following mixed strategy profile

$$\sigma_1 = \left[\tfrac{3}{10}(Al), \tfrac{1}{10}(Ar), \tfrac{4}{10}(Bl), \tfrac{2}{10}(Br)\right], \text{ and}$$

$$\sigma_2 = \left[\tfrac{1}{2}(L), \tfrac{1}{2}(R)\right].$$

- This strategy profile induces the probabilities of reaching all the nodes in the game.

Example

- Let's first consider x_5.
- This node can be reached if Player 1 plays A at x_0 then Player 2 plays L at x_1, and finally at the non-trivial information set of Player 1, l is played.
- This means Player 1 plays the strategy Al and Player 2 plays the strategy L. The probability of x_5 being reached is $\frac{3}{10} * \frac{1}{2} = \frac{3}{20}$.
- Similarly we can compute for other end nodes.
- For other vertices, note that, probability of x_3 being reached is the sum of the probabilities that x_5 and x_6 are reached (the descendants of x_3).

Example

- Similarly, given behavior strategies of both players, we can compute the probabilities of different nodes being reached.

Equivalence definition

Definition

A mixed strategy σ_i and a behavior strategy b_i of player i in an extensive form game are equivalent to each other if for every mixed/behavior strategy vector σ_{-i} of the players $N \setminus i$ and every vertex x in the game tree

$$\rho(x; \sigma_i, \sigma_{-i}) = \rho(x; b_i, \sigma_{-i}).$$

A reduced condition

- In the previous definition, it is enough to have the final condition hold for the end nodes in the game.
- This is because for any node x, the probability of x being reached is the sum of the probabilities that the end nodes that are descendants of x will be reached.

A result

Theorem

If a mixed strategy σ_i of Player i is equivalent to a behavior strategy b_i, then for every mixed/behavior strategy vector σ_{-i} of the other players and every player $j \in N$,

$$U_j(\sigma_i, \sigma_{-i}) = U_j(b_i, \sigma_{-i}).$$

Corollary

- Repeated application of the previous result yields the following corollary.

Corollary

Let $\sigma = (\sigma_i)_{i \in N}$ be a vector of mixed strategies. For each player i, let b_i be a behavior strategy that is equivalent to σ_i, and denote $b = (b_i)_{i \in N}$. Then, for each player i,

$$U_i(\sigma) = U_i(b).$$

Construction of a mixed strategy from a behavior strategies

- Given a behavioral strategy there is an obvious way to define an equivalent mixed strategy: for each pure strategy, simply multiply all probabilities assigned by the behavioral strategy to the actions occurring in the pure strategy.
- Formally, given b_i, we define a mixed strategy as

$$\sigma_i(s_i) := \prod_{h_i \in H_i} b_i(s_i(h_i), h_i)).$$

Example

- Consider the following game and consider the behavior strategy shown in the diagram.

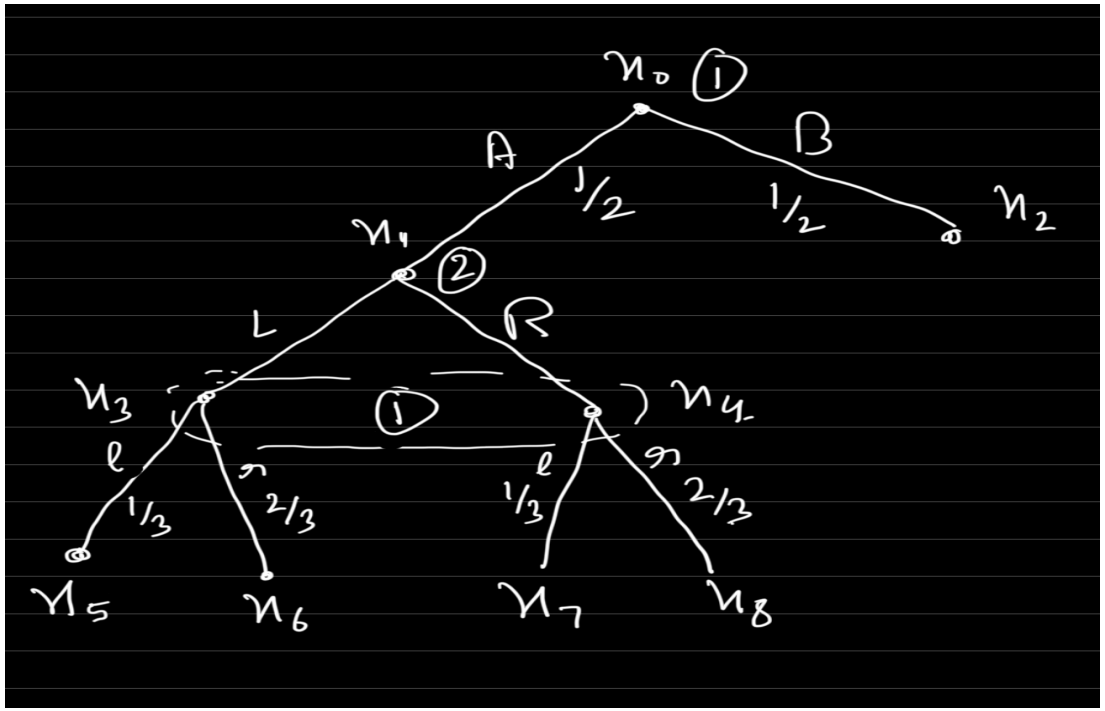

Figure: Figure 4

Example

- We can compute an equivalent mixed strategy in the following way:

$$\sigma_1(Al) = \frac{1}{2} * \frac{1}{3} = \frac{1}{6}$$

$$\sigma_1(Ar) = \frac{1}{2} * \frac{2}{3} = \frac{1}{3}$$

$$\sigma_1(Bl) = \frac{1}{2} * \frac{1}{3} = \frac{1}{6}, \text{ and}$$

$$\sigma_1(Br) = \frac{1}{2} * \frac{2}{3} = \frac{1}{3}.$$

Uniqueness

- It should be noted that there is not necessarily a unique mixed strategy that is equivalent to a given behavioral strategy.
- In the previous example, if we consider a behavior strategy which puts 0 probability to A, any mixed strategy that assigns 0 prob to Al and Ar is equivalent to the behavior strategy.

Result

Theorem

Let b_i be a behavioral strategy of Player i in Γ. Then there is a mixed strategy σ_i of Player i that is equivalent to b_i. Such a strategy σ_i is obtained by assigning to each pure strategy s_i of player i the product of the probabilities assigned by b_i to the actions chosen by s_i.